

Instructions: Answer all questions. Show working where applicable.

Q1. What is the speed of light in a vacuum? [2 marks]

Q2. Define the term 'refractive index' and write its formula. [4 marks]

Q3 (a). A ray of light travels from water ($n=1.33$) into glass ($n=1.5$).

If the angle of incidence is 30 degrees, find the angle of refraction using Snell's Law. [6 marks]

Q3 (b). State one real-world application of total internal reflection. [2 marks]

Q4. Draw a ray diagram showing how a concave mirror forms a real, inverted image. Label the principal focus and centre of curvature. [6 marks]